

Performance Testing

Date	30 June 2026
Team ID	SWTID-2026-9068
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing, Stress Testing
Target Module / API	/predict Loan Prediction API Endpoint
Test Environment	Local Flask Server (localhost:5050)
Test Date	30 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits loan application	10	60	Loan eligibility predicted successfully
2	Multiple users submit loan applications simultaneously	20	60	System handles concurrent requests without failure
3	Continuous loan prediction requests	30	120	Response time remains acceptable

4	Peak load with multiple loan applications	50	180	API remains stable and predictions are generated correctly
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Prediction generated quickly
2	Response Time (Max)	< 5 seconds	3.6 seconds	Pass	Slight delay during heavy load
3	Throughput (Req/sec)	10 Req/sec	13 Req/sec	Pass	Successfully handled expected traffic
4	Error Rate	< 1%	0.3%	Pass	Very few timeout errors
5	CPU Utilization	< 80%	61%	Pass	Stable CPU usage
6	Memory Utilization	< 80%	55%	Pass	Memory usage remained within limits

Step 4: Observations & Analysis

- The **/predict API** successfully handled multiple loan prediction requests simultaneously.
- The Random Forest Classifier generated loan approval predictions accurately under normal and peak load conditions.
- Average response time remained below the target value.
- The Flask backend remained stable without crashes during testing.
- CPU and memory utilization stayed within acceptable limits.

- The application successfully processed applicant information including income, education, credit history, and loan amount.
- Overall, the Smart Lender system demonstrated reliable performance and stable prediction accuracy during load testing.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

